



අධ්‍යයන පොදු සහතික පත්‍ර (උසස් පෙළ) විභාගය (ආදර්ශ), 2024

General Certificate of Education (Adv. Level) Examination (Model), 2024

|                |   |
|----------------|---|
| භෞතික විද්‍යාව | I |
| Physics        | I |

01 E I

පැය දෙකයි  
*Two hours*

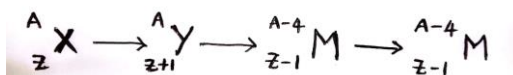
## Instructions

- Answer all the questions.
- Write your index number in the space provided in the answer sheet.
- Read the instructions given on the back of the answer sheet carefully.
- In each of the questions 1 to 50, pick one of the alternatives from (1), (2), (3), (4), (5) which is correct or most appropriate and mark your response on the answer sheet with a cross (x) in accordance with the instruction given on the back of the answer sheet. Use of calculators is not allowed.

$(g = 10 \text{ ms}^{-2})$

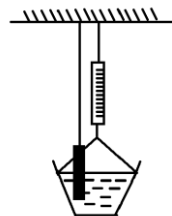
- Which of the following statements are incorrect regarding measurement errors?
  - The error in the measurement taken by an instrument calibrated falsely is considered to be a random error.
  - Zero error is a systematic error
  - A systematic error occurs when time is measured from a stopwatch whose time runs slowly.
  - The random error of a measurement can be decreased when it is taken multiple times
  - An increased proficiency in handling equipment of the person who is engaging in the experiment, contributes to a lower random error.
- There's a small box of mass 20kg on top of a box of mass 30kg resting on a smooth horizontal plane. The coefficient of friction between the boxes is 0.5. What is the acceleration of the system when a force of 50N is applied on the lower box?
  - $0 \text{ ms}^{-1}$
  - $1 \text{ ms}^{-1}$
  - $2 \text{ ms}^{-1}$
  - $3 \text{ ms}^{-1}$
  - $5 \text{ ms}^{-1}$
- What is observable when 25 is viewed through a laboratory hand lens?

(1) 25    (2) 52    (3) 57    (4) 52    (5) 75
- What are the radiations emitted respectively when the following radioactive chain reaction takes place?



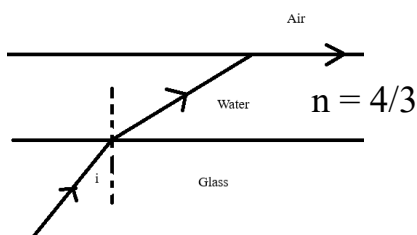
1.  $\alpha, \beta, \gamma$
  2.  $\alpha, \gamma, \beta$
  3.  $\beta, \gamma, \alpha$
  4.  $\beta, \alpha, \gamma$
  5.  $\gamma, \alpha, \beta$
5. When a water body is hung from a spring balance, the reading was 10kg. What is the reading of the spring balance when an iron block of mass 7.2kg is suspended from a string so that only half of its volume is submerged in water? (Relative Density of iron is 7.2 )

1. 10 kg
2. 10.5 kg
3. 12 kg
4. 13.6 kg
5. 17.2 kg



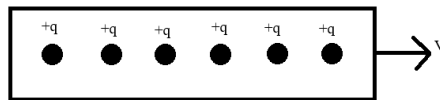
6. A ray indenting on glass-water surface with an angle of incidence  $i$ , emerges parallel to the water surface. Therefractive index of glass  $n_g$  is?

1.  $(4/3) \sin i$
2.  $1/(\sin i)$
3.  $4/3$
4.  $3/2$
5.  $9/8$



7. There is a spherical cavity inside a steel block of linear expansivity  $1.2 \times 10^{-5} \text{ } ^\circ\text{C}^{-1}$ . The percentage change of volume when the temperature increases by  $100^\circ\text{C}$  is?
1. increases by 0.12%.
  2. increases by 0.24%.
  3. increases by 0.36%.
  4. decreases by 0.36%.
  5. decreases by 0.12%.
8. A drill of power  $P$  is used to drill a hole in a brass box of mass  $M(\text{kg})$ . The specific heat capacity of brass is  $S (\text{J kg}^{-1} \text{ } ^\circ\text{C}^{-1})$ . There is a power loss of 40% due to heating of the drill. The increases of temperature in the box during  $T(\text{s})$  time?
1.  $(0.6 PT) / MS$
  2.  $(0.6 P) / MST$
  3.  $(0.4 PT) / MS$
  4.  $(0.4 P) / MST$
  5.  $PT / MS$
9. A child walks horizontally with  $2\text{ms}^{-1}$  velocity when rain drops travel vertically downwards at  $2\sqrt{3} \text{ ms}^{-1}$  velocity. If the child is not to be wet, what is the angle (respect to the horizontal) of his umbrella?
1.  $90^\circ$
  2.  $60^\circ$
  3.  $30^\circ$
  4.  $45^\circ$
  5.  $0^\circ$
10. The weight of an object on the Earth's surface is  $w$ . What is the weight of the object on the surface of a planet whose radius is half and mean density is twice that of earth?
1.  $w$
  2.  $2w$
  3.  $4w$
  4.  $8w$
  5.  $16w$
11. A person cannot see objects which are more than 2m away. An eye specialist prescribes a spectacle with a fixed lens contacted with a convex lens of focal length of 40cm. The optical power of the concave lens in diopter is.
1. +2
  2. +1.5
  3. -2
  4. +3
  5. -3

12. A liquid of density  $\rho_1$  and mass  $m_1$  is mixed with a liquid of density  $\rho_2$  and mass  $m_2$ . If the volume of the mixture does not decrease, what is the density of the mixture?
1.  $(\rho_1 - \rho_2) / 2$
  2.  $(\rho_2 - \rho_1) / 2$
  3.  $(\rho_1 + \rho_2) / 2$
  4.  $(2 \rho_1 + \rho_2) / 2$
  5.  $(\rho_1 + 2 \rho_2) / 2$
13.  $q$  charges are placed with  $x$  distance on an insulating sheet. If the sheet moves with  $v$  velocity, what is the mean current flowing along it?

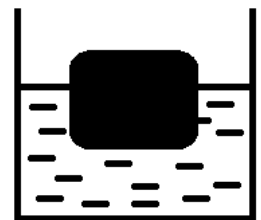


1.  $qv^2/x$
  2.  $qv/x^2$
  3.  $qv/x$
  4.  $q^2v^2/x$
  5.  $qv$
14. Consider the following statements made regarding the atmosphere.
- A. The relative humidity of dry atmosphere will never be 100%.
  - B. When relative humidity decreases absolute humidity also decreases in the atmosphere
  - C. Whenever the absolute humidity of atmosphere at any temperature is at its maximum, the relative humidity is 100%

Correct statements are?

1. B only
  2. C only
  3. A and B only
  4. A and C only
  5. B and C only
15. The ratio of turns in a step-down transformer is 20: 1. If it works with an efficiency of 80%, the ratio of primary current: secondary current is,
1. 1/25
  2. 1/20
  3. 1/16
  4. 16
  5. 25

16. A wooden box of volume  $0.04\text{m}^3$  is floating on water such that 75% of its volume is submerged in water. What is the minimum vertical force that should be applied on the box so that it completely sinks? (Density of water is  $1000\text{kgm}^{-3}$ )



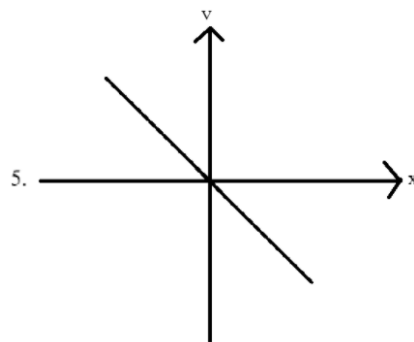
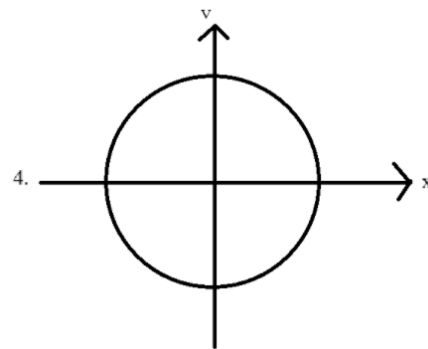
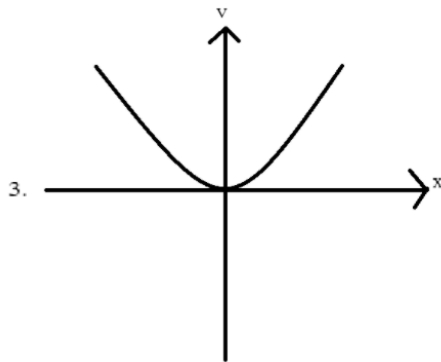
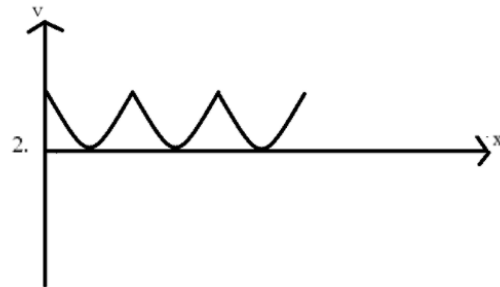
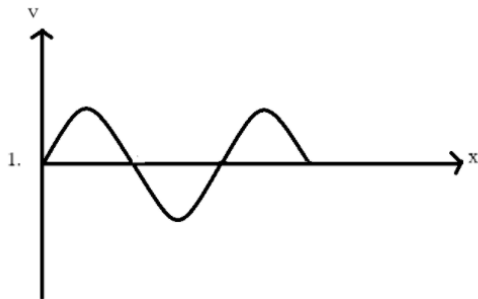
1. 10 N
  2. 30 N
  3. 100 N
  4. 300 N
  5. 400 N
17. What is the wave property which cannot be demonstrated using the ripple tank?
1. Reflection
  2. Refraction
  3. Diffraction
  4. Polarization
  5. Interference
18. When a certain capillary tube is inserted into water, the capillary rise of it is  $h$ . The angle of contact between water and glass is zero. The angle of contact of another tube with same dimensions is  $60^\circ$ . What is the capillary rise of water at that time?

1. 0
2.  $h/4$
3.  $h/2$
4.  $h$
5.  $2h$

19. What is the electric field intensity near an infinitely large sheet of surface charge density of  $\sigma$ ,

1.  $\sigma / \epsilon_0$  Parallel to sheet
2.  $\sigma / \epsilon_0$  Perpendicular to sheet
3.  $2 \sigma / \epsilon_0$  Perpendicular to sheet
4.  $\sigma / 2\epsilon_0$  Parallel to sheet
5.  $\sigma / 2\epsilon_0$  Perpendicular to sheet

20. What is the velocity(v) - displacement(x) graph of an object engaging in simple harmonic motion?

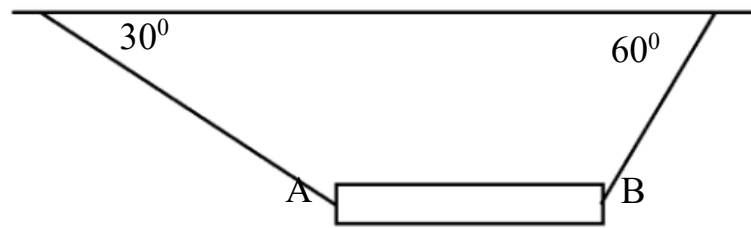


21. The method to increase the sensitivity of a moving-coil galvanometer is,

- A. Increasing the number of turns in the coil
- B. decreasing the strength of magnetic poles
- C. By winding the coil in a iron cylinder with a much larger diameter.
- D. by attaching to a hair spring of Higher torsion constant

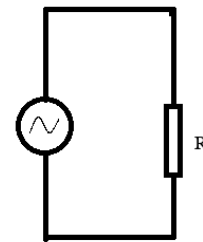
1. A and B only
2. A and D only
3. A and C only
4. A, B and C only
5. B, C and D only

22. The rod AB is supported by 2 strings as shown below. If the center of gravity of the rod is G, then the ratio AG/GB is,



1. 3:1
  2.  $\sqrt{3}:1$
  3.  $1:\sqrt{3}$
  4. 1:1
  5. 1:3
23. The maximum value of alternative current supplied by the alternative power source is  $I_0$ . What is the mean power dissipation of the resistor R?

1.  $(1/2) I_0^2 R$
2.  $I_0^2 R / \sqrt{2}$
3.  $I_0^2 R$
4.  $\sqrt{2} I_0^2 R$
5.  $2 I_0^2 R$



24. The Mercury thermometer is calibrated such that the gap between two adjacent temperature marks is 1mm. What is the distance between two adjacent temperature marks if the volume of the bulb of the thermometer is doubled while the radius of the capillary tube is halved?

1.  $\frac{1}{4}$  mm
2.  $\frac{1}{2}$  mm
3. 2 mm
4. 4 mm
5. 8 mm

25. Consider the following statements regarding a transistor

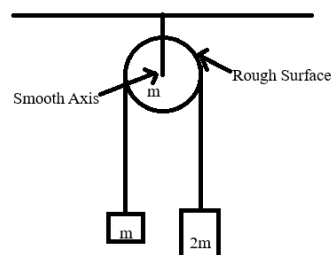
- A. When B-E junction is forward biased and B-C junction is reverse biased, the transistor can be used as an electric signal amplifier.
- B. When  $V_{ce} = V_{cc}/2$  the input electric signal of a transistor is amplified without distortion and obtained as output.
- C. When a transistor biased towards saturated region  $V_{ce} \sim 0$

Which is true?

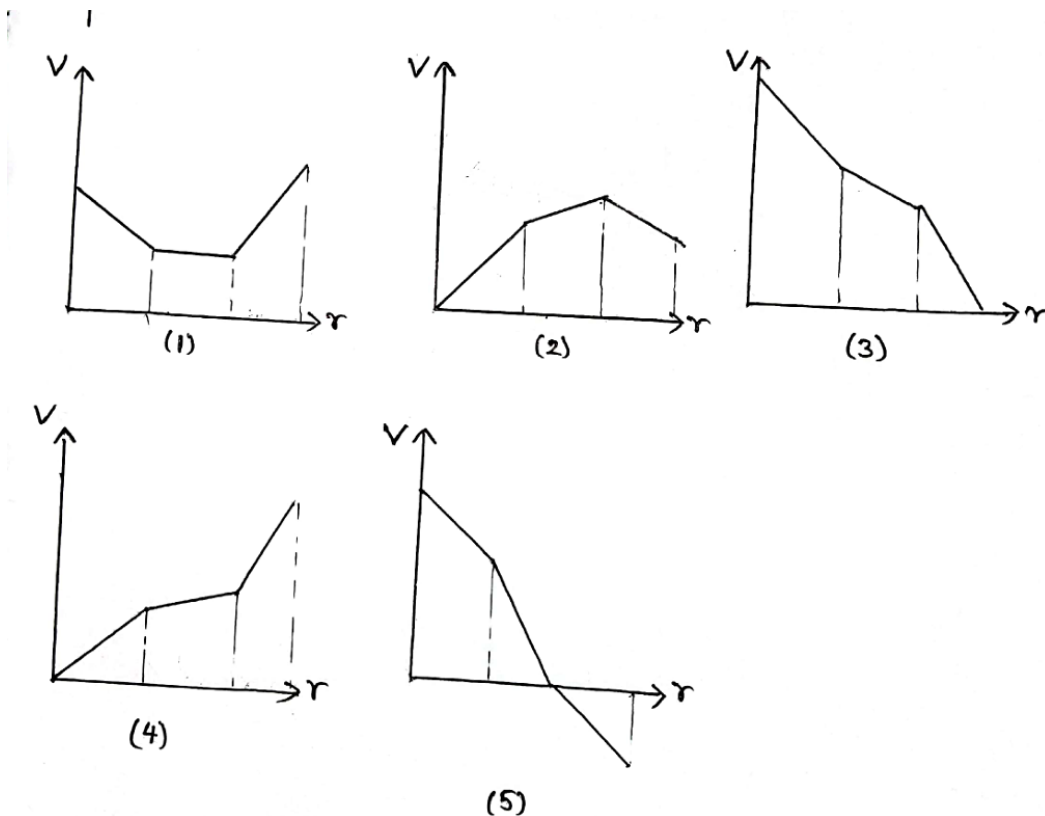
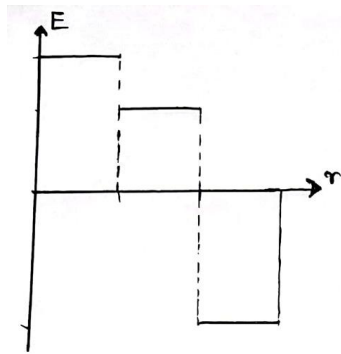
1. A
2. A, B
3. B, C
4. A, C
5. A, B, C

26. As shown in the figure, masses  $m$  and  $2m$  are hung on two sides of a disc like pulley of mass  $m$ . When it is gently released, what is the acceleration of  $2m$ ? (The moment of inertia of a disc of mass  $m$  and radius  $r$  is  $(\frac{1}{2})mr^2$ )

1.  $2g/7$
2.  $4g/7$
3.  $g/5$
4.  $g/4$
5.  $g$



27. The body of mass  $m$  is kept on a rod which rests on a horizontal smooth plane. The coefficient of friction between the body and the rod is  $\mu$ . The rod engages in a simple harmonic motion in the  $x$  axis with a time period of  $T$ . What is the maximum amplitude of rod such that the body does not slip on the rod?
1.  $\mu g T^2 / 4\pi^2$
  2.  $\mu g T^2 / 4\pi$
  3.  $2\mu g / T^2$
  4.  $4\mu g / T\pi$
  5.  $2\mu g / T\pi^2$
28. A small ball of mass  $M$  and density  $d_1$  is released into a tall beaker filled with glycerin of density  $d_2$ . After some time in the glycerin, the ball achieved constant velocity. What is the viscous force acting on the ball?
1.  $Md_1g/d_2$
  2.  $Mg(1 - d_2/d_1)$
  3.  $M(d_1 + d_2)/g$
  4.  $Md_1d_2$
  5.  $Mg\{(d_2/d_1) - 1\}$
29. The graph given below denotes the variation of electric field intensity ( $E$ ) and distance ( $r$ ) of an electric field. What is the correct graph which corresponds to the variation of potential ( $v$ ) and distance ( $r$ )?



30. A fluid enters a capillary tube of length 30cm with a pressure of  $1.05 \times 10^5$  Pa and leaves it at a pressure of  $1 \times 10^5$  Pa. What is the pressure at a point in the tube at a distance of 10cm from entry?

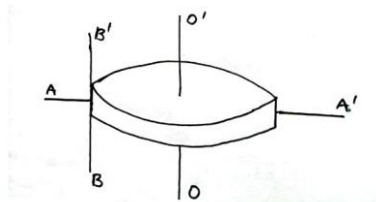
1.  $1.0 \times 10^5$  Pa
2.  $1.02 \times 10^5$  Pa
3.  $1.01 \times 10^5$  Pa
4.  $1.03 \times 10^5$  Pa
5.  $1.04 \times 10^5$  Pa

31. A boat travels at a uniform velocity of  $10\text{ms}^{-1}$ . The resistive force against the motion of the boat is 400N. What is the power of the engine?

1. 40 W
2. 400 W
3. 2000 W
4. 3000 W
5. 4000 W

32. If the moment of inertia of a thin, uniform disc about the axes  $OO'$ ,  $AA'$  and,  $BB'$  is  $I_1$ ,  $I_2$ , and  $I_3$ ,

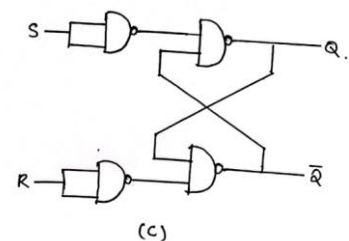
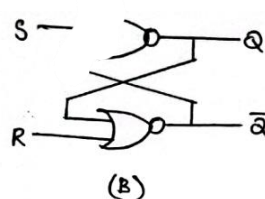
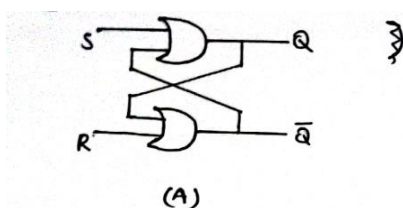
1.  $I_1 > I_3 > I_2$
2.  $I_3 > I_1 > I_2$
3.  $I_1 > I_2 > I_3$
4.  $I_2 > I_3 > I_1$
5.  $I_2 > I_1 > I_3$



33. A resonance tube with end correction  $e$  is completely immersed in water and is slowly lifted. The length difference between the air columns of the first two resonance instances is  $L$ . If the speed of sound in air is  $V$ , what is the frequency of the tuning fork?

1.  $V / 2(L + e)$
2.  $V / (2L + e)$
3.  $V / (L + e)$
4.  $V / L$
5.  $V / 2L$

34. Which of the following logic circuits corresponds to S-R Flip flops?

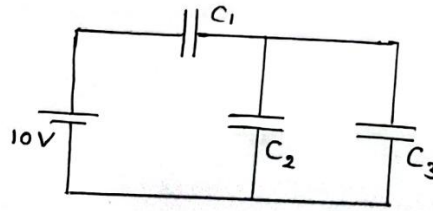


1. A only
2. B only
3. A and C only
4. B and C only
5. A, B and C all

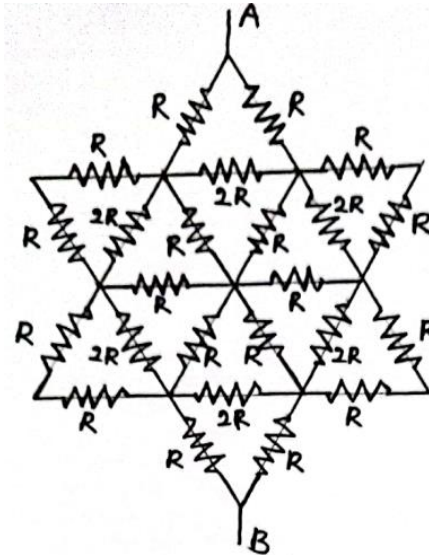
35. The cell given in the following circuit has no internal resistance and an EMF of 10V. What is the charge and potential difference of  $C_1$  capacitor?

( $C_1 = 6 \mu\text{F}$ ,  $C_2 = 9 \mu\text{F}$ ,  $C_3 = 3 \mu\text{F}$ )

1.  $12.6 \mu\text{C}$ ,  $3.67 \text{ V}$
2.  $20 \mu\text{C}$ ,  $5.33 \text{ V}$
3.  $30 \mu\text{C}$ ,  $6.67 \text{ V}$
4.  $40 \mu\text{C}$ ,  $6.67 \text{ V}$
5.  $90 \mu\text{C}$ ,  $7.33 \text{ V}$

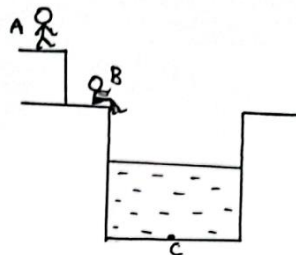


36. What is the equivalent resistance between A and B?



1.  $R/2$
2.  $R$
3.  $3R/2$
4.  $2R$
5.  $5R/2$

37. Two people, A and B, are looking at a bulb, B, at the bottom of a tank filled with water, as shown in the figure below. Person B is sitting, and person A is standing behind him. The correct statement regarding the visual position of the bulb as seen by A and B is,

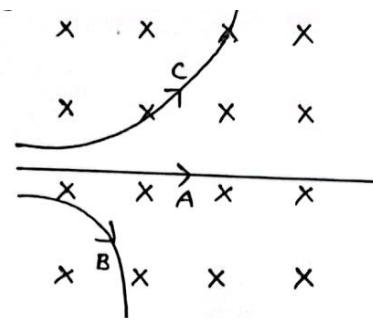


1. The position of the bulb as seen by person B is higher than the position as seen by person A.
2. The position of the bulb as seen by person B is lower than the position as seen by person A.
3. The visual position from which the bulb is visible to both persons A and B is the same.
4. Person B can see the bulb, but person A cannot see the bulb.
5. Person A can see the bulb, but person B cannot see the bulb.



38. The following figure shows the three paths of three particle which moving in a magnetic field. If those paths are A , B and C the respective particles could be,

|    | A | B | C             |
|----|---|---|---------------|
| 1. | n | p | e             |
| 2. | n | e | $\gamma$ rays |
| 3. | n | e | p             |
| 4. | e | p | n             |
| 5. | p | e | n             |



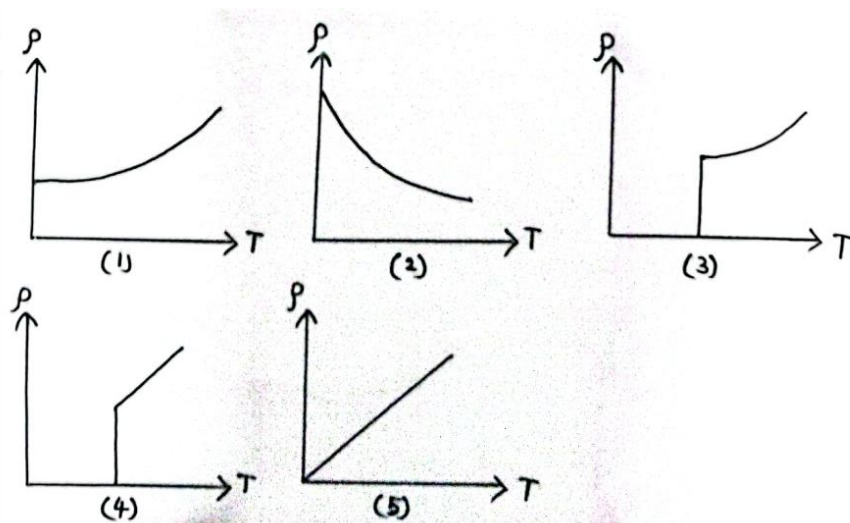
39. What is the phenomena that cannot be explained using the Bernoulli principle?

1. When a trainee is moving fast, a person who is near the train track being pulled towards it.
2. Increasing the speed of water when flowing from large tube to a smaller tube.
3. Breaking off and falling of low weight sheets from the roof of a house due to a high wind.
4. Sinking of near ground water places due to flow of water through tunnels.
5. Ejaculation of water from a pump (injector) due to compression.

40. Frequency of a string which is tied up at both ends is thousand hertz if the tension of the string is increased by 21% the new frequency is,

1. 1021 Hz
2. 1100 Hz
3. 1210Hz
4. 420 Hz
5. 790 Hz

41. For a superconductor the variance between its absolute temperature(T) and resistivity( $\rho$ ) is well depicted by,



42. Two identical capacitors have the same capacitance C, one is charged to a voltage  $V_1$  and the other to a voltage  $V_2$ . Then, the energy loss in the system connected with their negative plates together and their positive plates together is,

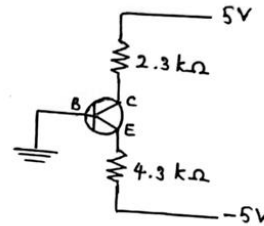
1.  $C (V_1^2 - V_2^2) / 4$
2.  $C (V_1^2 + V_2^2) / 4$
3.  $C (V_1 - V_2)^2 / 4$
4.  $C (V_1 - V_2)^2 / 2$
5.  $C (V_1 + V_2)^2 / 4$

43. Radioactive element  $\alpha$  is turned into a stable element  $\gamma$  if the half-life of the  $\alpha$  is 8 days find the percentage of  $\alpha$  atoms which is turned into  $\gamma$  after 24 days,

1. 25%
2. 50%
3. 75%
4. 87.5%
5. 95.75%

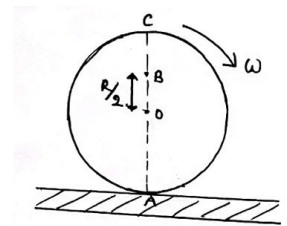
44. The following figure depicts a circuit with a silicon npn transistor the value of  $V_{CE}$  nearly is

1. 2.7 V
2. 3.3 V
3. 3.7 V
4. 4.3 V
5. 4.7 V



45. A uniform disc with radius  $R$  is rotating with an angular velocity. It is placed on a smooth horizontal plane freely.  $V_A$ ,  $V_B$  and  $V_C$  are linear speeds of points A, B and C, what is correct?

1.  $V_A = V_B = V_C$
2.  $V_A = V_B > V_C$
3.  $V_A = V_C > V_B$
4.  $V_A > V_B = V_C$
5.  $V_A < V_B < V_C$



46. Consider the state in which water turns into steam according to dynamics equations  $\Delta Q = \Delta U + \Delta W$ , during the change what is true?

- A.  $\Delta U = 0$
- B.  $\Delta W > 0$
- C.  $\Delta Q > 0$

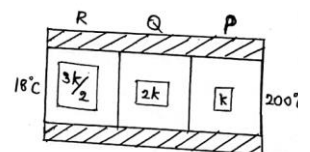
1. A only
2. A and B
3. A and C
4. B and C
5. A, B, C all

47. If the temperature of a black body is reduced by 50% the rate of radiation of energy reducing factor is,

1. 1/4
2. 1/16
3. 1/18
4. 1/2
5. 1/32

48. Three blocks P, Q and R with same dimensions are kept as shown in the figure and their surfaces are covered. The P and R ends are maintained at temperatures  $200^\circ\text{C}$  and  $18^\circ\text{C}$ . If the thermal conductivities of those three are  $K$ ,  $2K$  and  $3K/2$  respectively. After reaching the steady state the temperature difference between two ends of P is.

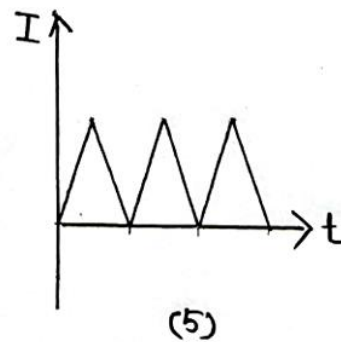
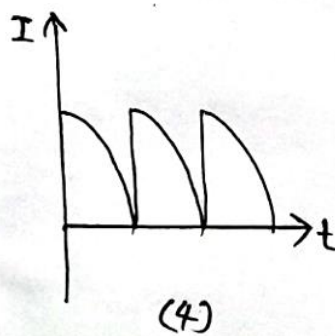
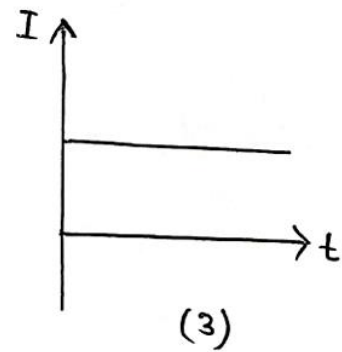
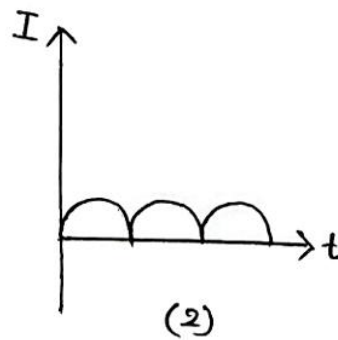
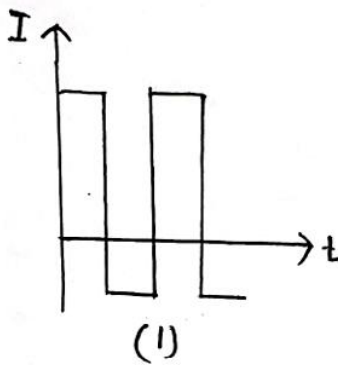
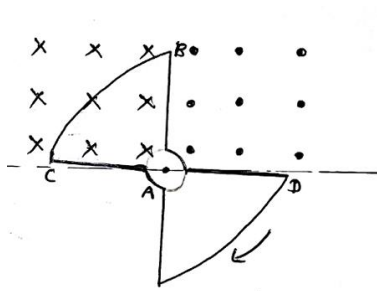
1.  $150^\circ\text{C}$
2.  $126^\circ\text{C}$
3.  $84^\circ\text{C}$
4.  $52^\circ\text{C}$
5.  $44^\circ\text{C}$



49. An object starts from rest and travels with uniform acceleration along a smooth horizontal plane and reached a velocity of  $V$  after  $n$  seconds. The displacement of the object between the time duration of  $(n - 2)$  and  $n$  is

1.  $2nV / (n + 1)$
2.  $2nV / (n - 1)$
3.  $2V / n$
4.  $2V(n - 1) / n$
5.  $V(n^2 - 1) / 2n$

50. The conducting loop shown in the figure rotates with a uniform angular velocity around its axis A. There are two uniform magnetic fields acting along the plane which are to the place that the loop rotates the induced current in the loop varies with time  $T$  is well depicted by.





- b) Suggest a method to determine the inner radius of the capillary tube without using mercury?
- .....
- .....
- .....
- .....
- .....
- c) What is the fact that should consider about the capillary tube while changing  $h$  in the experimenting time?
- .....
- d) What is the equipment named Q and list its advantages?
- .....
- .....

## II.

- a) Write down an expression for the pressure at the end A of the U tube if the liquid which is used highly is water and its density is  $\rho_0$  and atmospheric pressure is  $P$ ,
- .....
- b) Write down an expression for the pressure at the end B,
- .....
- c) If the radius of the capillary tube is  $r$  and the length of the capillary tube is  $l$  and if the water volume that is added to the beaker within  $t$  time is  $V$  by properly determining the dependent and independent variable correctly, build an expression in the form of  $y = mx$  in order to determine the coefficient of viscosity in graphical format,
- .....
- .....
- .....
- .....
- .....
- d) Find the coefficient of viscosity of water if the gradient of the graph is  $1 \times 10^{-4} \text{ m}^2\text{s}^{-1}$  and the radius of the capillary tube is  $0.2\text{mm}$  and the density of water is  $1000 \text{ kgm}^{-3}$  and the acceleration due to gravity is  $10 \text{ ms}^{-2}$  and length of the capillary tube is  $60 \text{ cm}$ ,
- .....
- .....
- .....
- .....

## III.

- a) If the apparatus is fixed as above during the experiment which liquid is initially filled into the U – tube? (density of bromine  $\rho_1 \gg$  density of coconut oil  $\rho_2$ )
- .....

b) After filling the boat liquids what end of the U - tube is fixed to the capillary tube with the pressure valve?

.....

c) Why is it done?

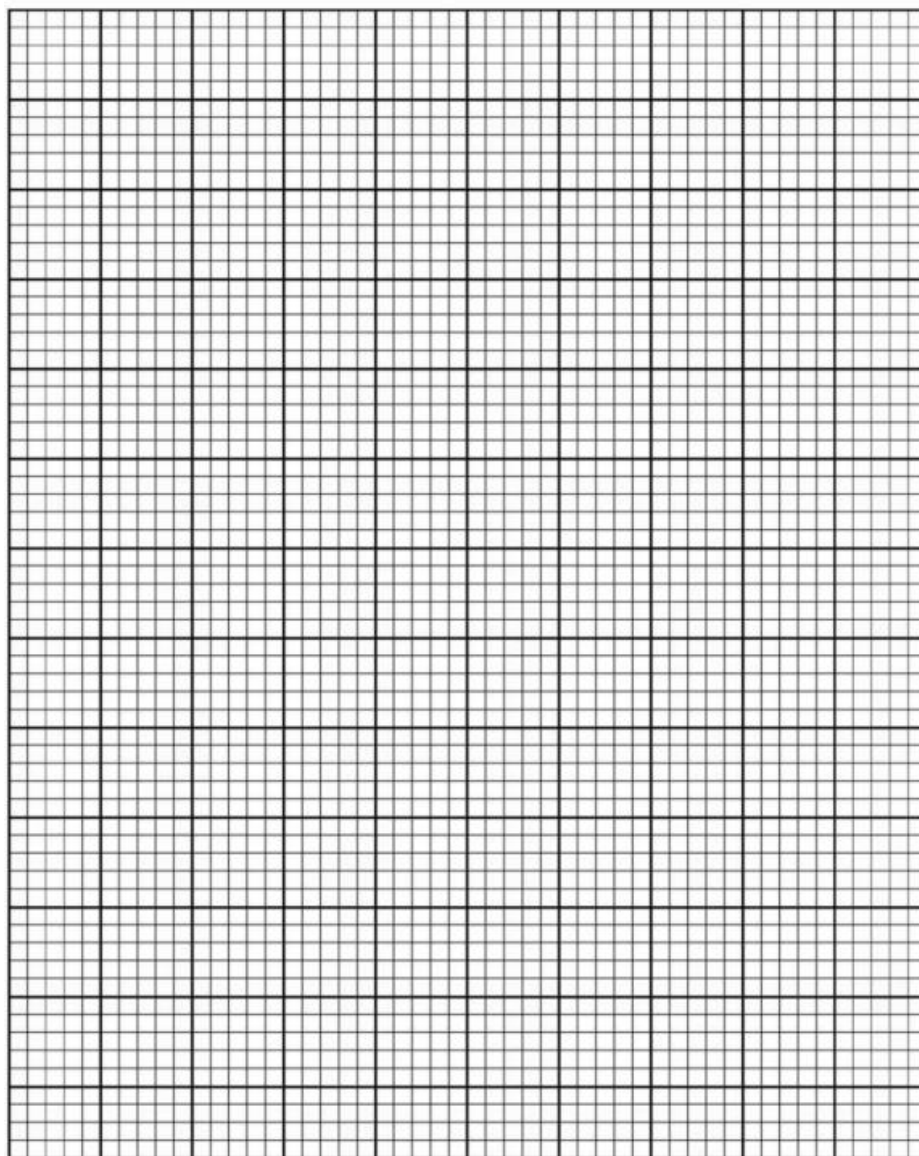
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d) A graph is obtained by measuring the height from interface of the open end ( $h_1$ ) by changing  $h$ . The readings are shown in the table below. Consider that the interface of the U - tube is always above the U bend. The height of the coconut oil is 10 cm. Input the given data in the below graph.

| $h$   | $h_1$   |
|-------|---------|
| 25 cm | 11 cm   |
| 35 cm | 14.3 cm |
| 45 cm | 17.6 cm |
| 55 cm | 21 cm   |
| 65 cm | 24.3 cm |



e) Hence find the densities of bromine and coconut oil.

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IV. Why do we not take high values for h? If we take high values for h, then draw the graph that is obtained.

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2. A student wants to determine the specific heat capacity of a liquid by using method of mixtures. For that calorimeter, sufficient amount of the liquid, small metal balls, metal powder, metal blocks, water, a water boiler, normal test tubes, Pyrex test tubes, thermometers, insulation cover, tripod, Bunsen burner, wire gauge and stirrer are supplied.

a) Initially the specific heat capacity of the metal should be found for that what format of the metal is used? Give two reasons for why. Also give two reasons respectively why doesn't other formats are not used.

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b) In here 2 thermometers are willing to use, state the ranges of those two thermometers and give reasons for choosing those ranges.

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c) State the methodology which is used to heat the metals and draw its apparatus.

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d) What are the two main factors that must be considered when adding heated metal into the calorimeter?

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e) What is the water amount that should be added to the calorimeter and why is that?

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f) What are the steps that should be considered after adding metal into the calorimeter with water?

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g) Below results are obtained after the completion of the whole experiment.

|                                                                |                           |
|----------------------------------------------------------------|---------------------------|
| Mass of the calorimeter with the stirrer                       | 250 g                     |
| Mass of the water used                                         | 150 g                     |
| Specific heat capacity of calorimeter                          | $450 \text{ J kg K}^{-1}$ |
| Mass of the liquid which the specific heat capacity is unknown | 150 g                     |
| Mass of the metal used                                         | 200 g                     |





i. What is the measuring instrument used?

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ii. If the zero error of that instrument is 0.3mm and diameters of those five wires are 0.5 mm 1.1 mm, 1.4 mm, 1.9 mm and 2.1mm respectively. Find the mass of unit length of each wire.

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iii. State how a more accurate reading is obtained from the instrument stated in part (i.),

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b) State how the experiment is carried out using each wire.

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c) The lengths obtained by these are 16 cm, 14cm, 11.5 cm, 9.7 cm, 8.3 cm. Build a graph with this data.

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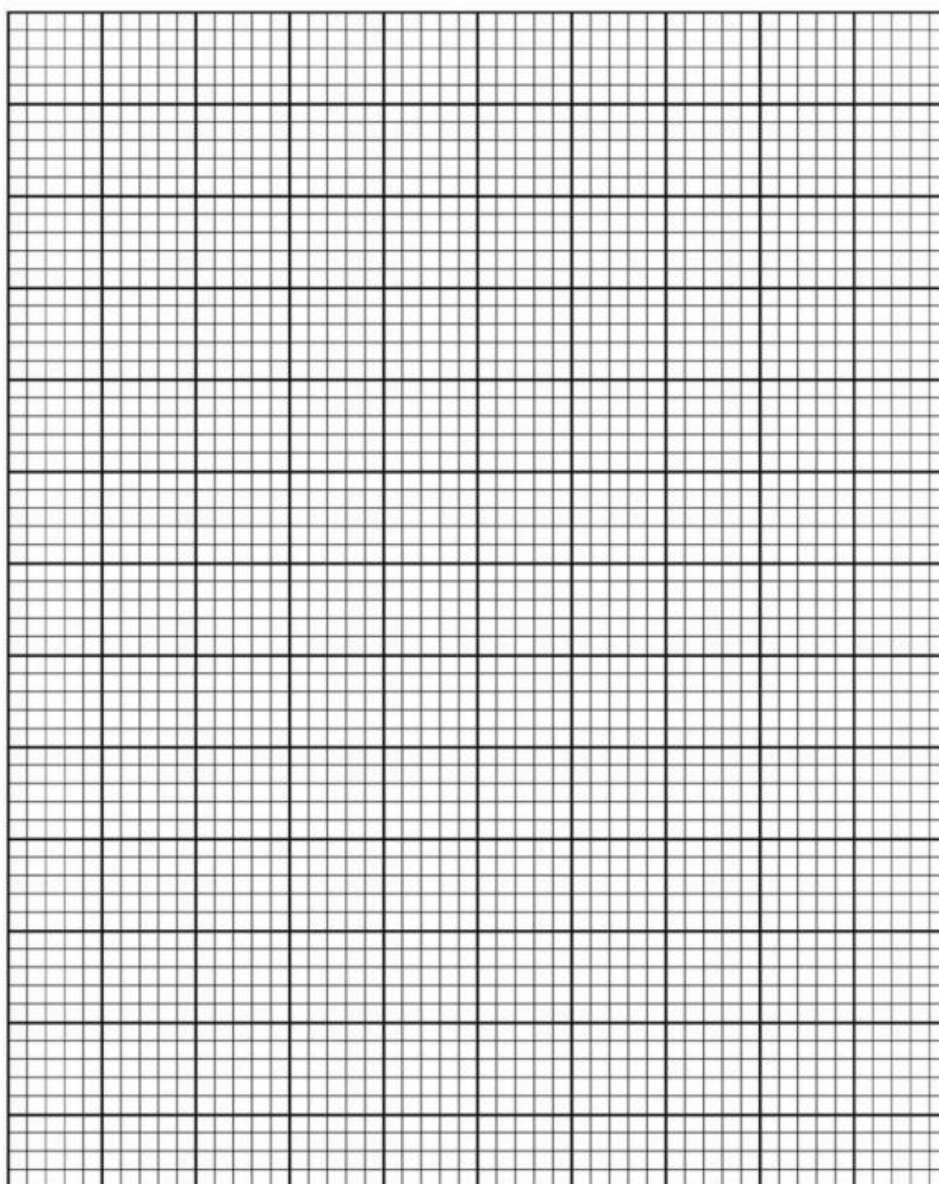
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d) Find the frequency of the tuning fork by using the graph.

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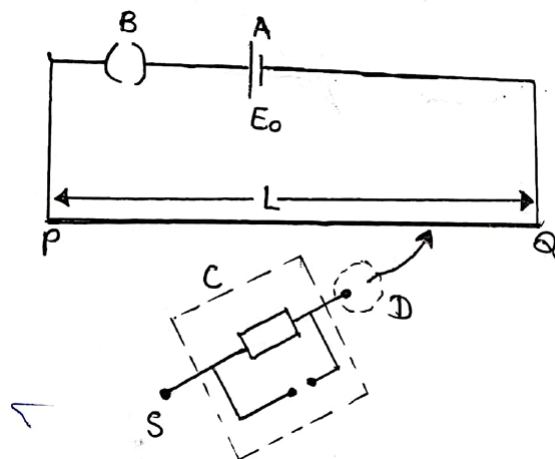
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- e) While using the tuning fork the student dropped it many times and its nature has been changed, hence draw the graph obtained in the same graph. (In here the student dropped the tuning fork before taking the reading of each wire and on each occasion the nature of the tuning fork changes) Mark the graph as x.
- f) Another student adds a drop of wax to the tuning fork before every occasion. Draw the graph hence obtained in the same graph. Now mark it as y.
- g) Another student, without changing the nature of the tuning fork uses human bones found in the biology lab as the loads to increase the tension in the sonometer wire every time. Draw the graph hence obtained in the above graph and name it as z.

4. A circuit diagram of a potentiometer used in the laboratory is given below.



i. Name A, B, C, D

A - .....

B - .....

C - .....

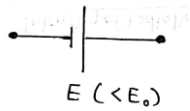
D - .....

ii. Name an essential item in A and D,

A - .....

D - .....

iii. Draw how the below dry cell connects to the potentiometer circuit to find its electromotive force.



iv. The above dry cell has  $E = 2\text{ V}$  and has a negligible internal resistance. Resistance of PQ is  $10\ \Omega$ . If  $L = 200\text{ cm}$ , find the potential drop per unit length.

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v. If the potentiometer can calibrate a  $4\text{ mV}$  heater, find the equilibrium length.

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vi. Are you satisfied with the above reading? Explain.

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vii. What is the change you make to the identified problem in question (vi)?

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viii. What is the resistance of the item used to maintain a 10 mV potential difference between the ends of potentiometer?

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ix. Find the new equilibrium length for the changed potentiometer when calibrating a heater.

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x. Compare the fractional errors of the equilibrium lengths took in both opportunities. Fractional errors for first and second equilibrium lengths are  $F_1$  and  $F_2$  respectively.

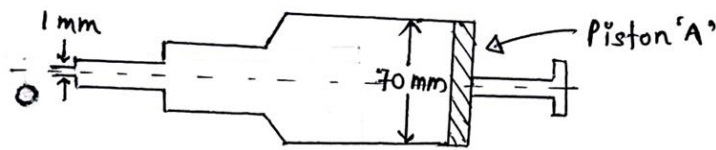
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**Part B – Essay**

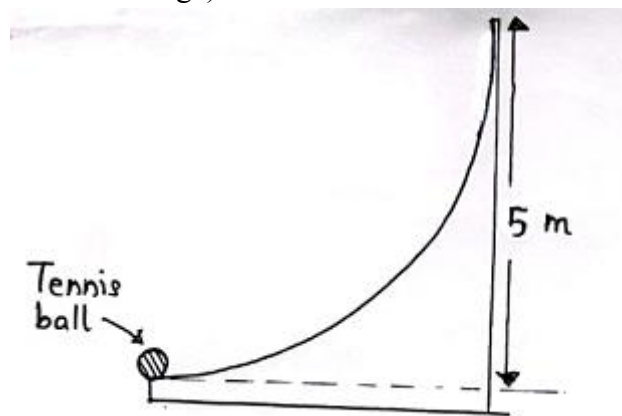
**Answer four questions only. ( $g = 10 \text{ ms}^{-2}$ )**

5.

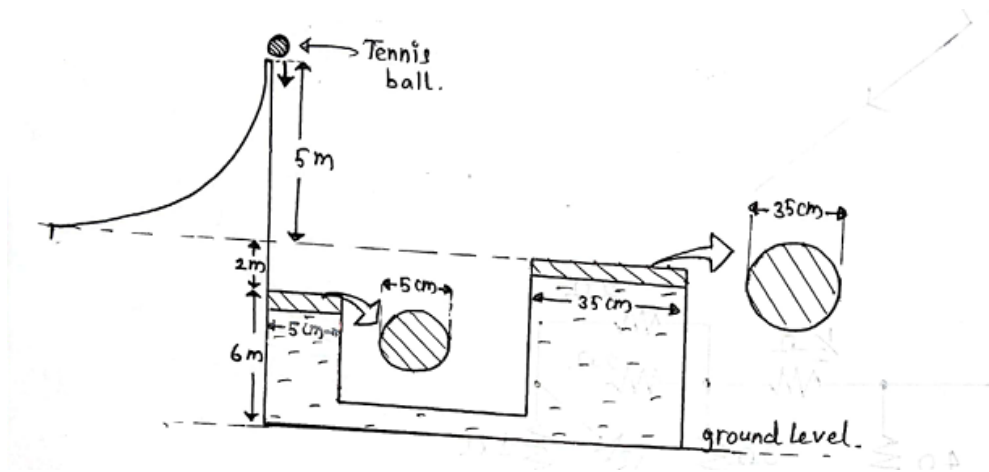
- i.
  - a. Define Bernoulli's principle.
  - b. The figure below shows a large syringe made using a PVC tube which is used as a toy water gun for children. The empty water gun is inserted into a bucket of water and water is stored into it by pulling the piston A. If a child wants to spray water column with the uniform speed of  $5 \text{ ms}^{-1}$ , at what speed should they push the piston?



- ii. The water column sprayed at the above speed is shot at a tennis ball with a mass of 20 g kept on a smooth plane as shown in the figure below. if the time of action of the force is 0.6 seconds and after the collision if the tennis ball reaches an unstable equilibrium with a moment stopped at the upper end of the plane. calculate the speed of the water column after hitting the tennis ball. (assume that the direction of water column does not change)



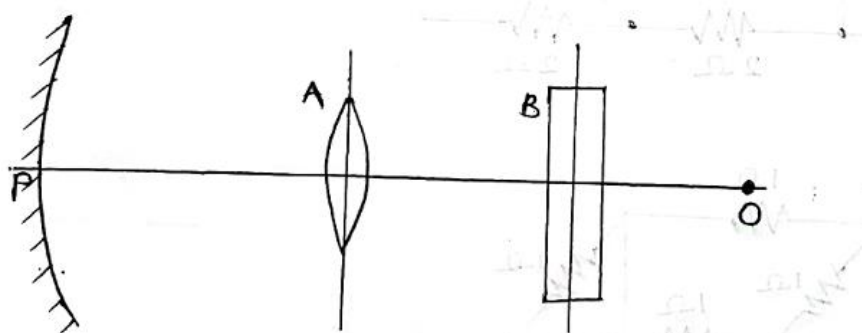
- iii. At the next moment the above-mentioned tennis ball at unstable equilibrium falls down out of the smooth plane and onto a pascal's law apparatus as shown in the figure below. in this collision of the tennis ball and the apparatus, the time of the action is 0.6 seconds. Calculate the maximum mass that can be lifted by the other terminal of the apparatus during the above time period if the ball moves 2 meters vertically upwards after the collision and assume that there is no energy loss during the process.



- iv. A child with the water gun mentioned in the question (i) shoots a straight column of water upwards at a  $45^\circ$  angle towards the tennis ball which is at a state of momentary rest after bouncing two meters up from the piston surface of the pascal's apparatus as mentioned in question (iii) above, the tennis ball then starts to move in a projectile motion with the same angle after hit by the water column shot with an uniform speed. it takes 2.5 seconds to hit the ground, ground level is same as in the question (iii) above and 0.5 seconds prior to it the ball reaches the same level of the height where it started the projectile motion. if the initial speed of water column is  $6 \text{ ms}^{-1}$  the collision time is one second and the water column moves in the same direction as before after the collision occurred calculate the new speed of water after the collision.

6.

- i.
  - a. Define the cartesian sign convention.
  - b. What is astigmatism a refractive error in the eye? What is the specific shape of the corrective lenses used to correct these errors?
- ii. Derive an expression for the focal length ( $f$ ) of a compound lens which consists of two convex lenses with focal lengths  $f_1$  and  $f_2$ .
- iii.

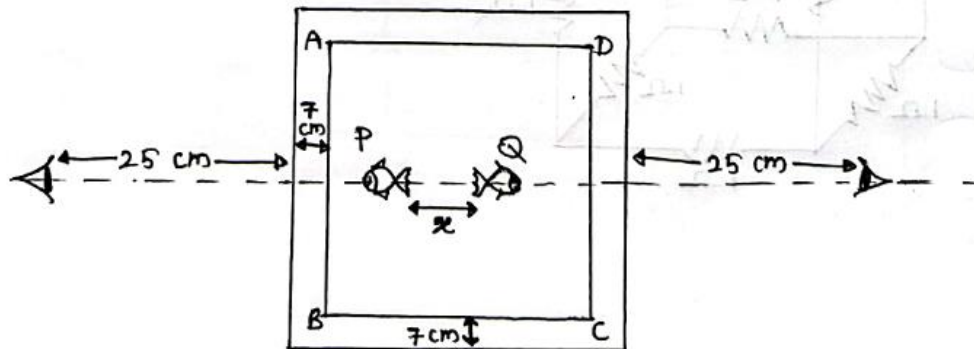


As shown in the above diagram a narrow laser beam emits from a laser torch at O, refracts through glass block B and convex lens A before it incidents on the concave mirror. if the laser beam reflects back from the mirror, travels parallel to the OP principal axis to strike on glass block B; only after refracting through the lens A, calculate the distance between the point O and the pole of the mirror P. radius of the curvature of the mirror is 35 cm, focal length of this convex lens 10 cm, the thickness of



the glass block is 7 cm, the distance between the lens A and the point O is 13 cm and the refractive index of glass is  $\frac{3}{2}$ .

- iv. As shown in the below figure a person who stands 25 cm away from the tank but with lone sightedness, looks at the two fish who are inside the tank with glass walls of uniform thickness of 7 cm from both sides AB and CD. The fish P looks 2.5 cm closer when he observes through side AB and the fish Q looks 3.1 cm closer when he observed through the side CD. if the two fish are identical and their length is 8 cm and in the same straight line, calculate the distance between P & Q (x) and the power of the eye lens. near point of this person is 45 cm away from the eye lens. (refractive index of water = 1.33 and refractive index of glass = 1.52)



- v. Students want to observe leaf mesophyll cells, but the light microscope of the school's biological laboratory has been broken. A student suggests making a temporary microscope to observe leaf mesophyll cells by using equipment in the physics laboratory. (Consider that they want to obtain the maximum magnitude possible.)

- a. What is the minimum number of lenses that should be used by students for the above process and what are those lens types?

- b. In the physics laboratory there are lenses with the following focal lengths,

+/- 2 cm

+/- 4 cm

+/- 0.5 cm

If the student chooses lenses, they need as in (a) above,

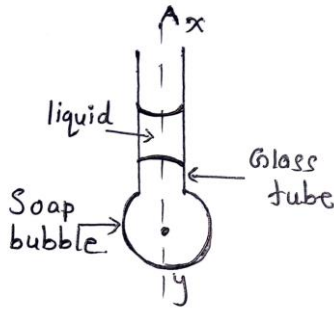
- i) Draw an accurate diagram to show how the students should place the lenses they choose, their specimen (leaf mesophyll cells) and the place where the final image is formed and make all the distances as relevant. (distance between the specimen and the objective lens is 2cm)
- ii) Calculate the angular magnification of the temporary microscope created by the student as mentioned above.

7.

i.

- a. What is known as cohesion?
- b. What is known as adhesion?
- c. Define coefficient of surface tension.

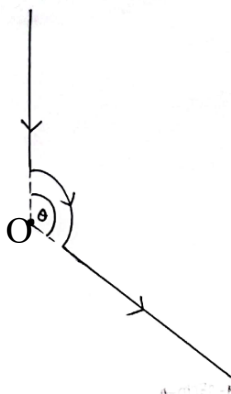
- ii. Draw the graph of pressure variation from x to y.



- iii. Derive the expression for the capillary rise using the expression which shows the pressure difference across the spherical liquid surface, in terms of density of the liquid, surface tension of the liquid, radius of the curvature of liquid meniscus and gravitational acceleration.
- iv. When capillary tube of radius  $3 \times 10^{-4}$  m immersed vertically in a liquid in a beaker its liquid rises to  $3.19 \times 10^{-2}$  m then the pressure in the capillary tube was increased, and it was measured using a pressure meter. at this point an air bubble occurred at the bottom of the tube and when the bubble is going to break the difference of the liquid columns of the manometer was  $5.11 \times 10^{-2}$  m. if the bottom of the capillary tube was kept  $2.37 \times 10^{-2}$  m and below the liquid surface and if the densities of the liquids in the beaker and manometer were  $790 \text{ kgm}^{-3}$  and  $1025 \text{ kgm}^{-3}$  respectively, find the surface tension of the liquid in the beaker and find the contact angle between the liquid and glass.

8.

- i.
- Define magnetic flux density.
  - Draw the magnetic field between the poles of a U-shaped magnet
  - Derive an expression for the magnetic flux density at the center of a conductive loop where current is passed according to Biot-Savart rule. Introduce all the terms used.
- ii. A cylindrical wooden block of length 7 cm and mass 0.45 kg is given in the figure below. A rectangular coil with 50 turns is coiled along the cylindrical wood across the diameter of the cylinder. The coil is placed at an  $\alpha$  angle inclined to the cylinder so that the coil is parallel to the inclined plane. Plane is smooth. find the current across the coil needed to prevent the cylinder from being slide downwards. consider that a uniform magnetic field of flux density 0.4 T acts vertically downwards at this point.
- iii. The radius of the circular part is R and  $\theta = 120^\circ$ . Current is I. Find the magnetic flux density at O.



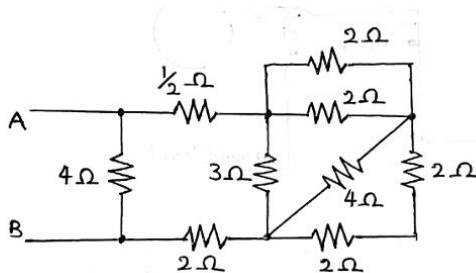
iv. There are  $3 \times 10^6$  turns in a unit length of a transformer. A coil of radius 6 cm with 50 turns is kept co-axially with the axis of the transformer. The current across the transformer decreases from +3 A to -3 A during a period of 0.045 s.

Find below,

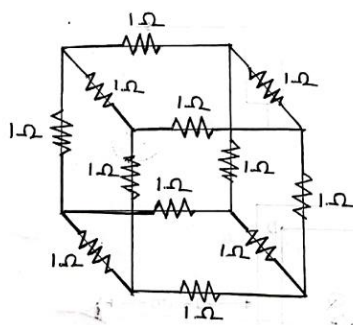
- The rate of change of current across the transformers.
- Rate the change of magnetic flux across the coil.
- Induced electromotive force in the coil.
- If the resistance of the coil is  $15 \Omega$ , the total charge passes during that time.  
( $\mu_0 = 4 \times 10^{-7} \text{ Hm}^{-1}$ )

## 9. Part A

- Define Kirchoff Laws.
  - Define current density.
- Find the resistance between A and B in the given circuits,



b.

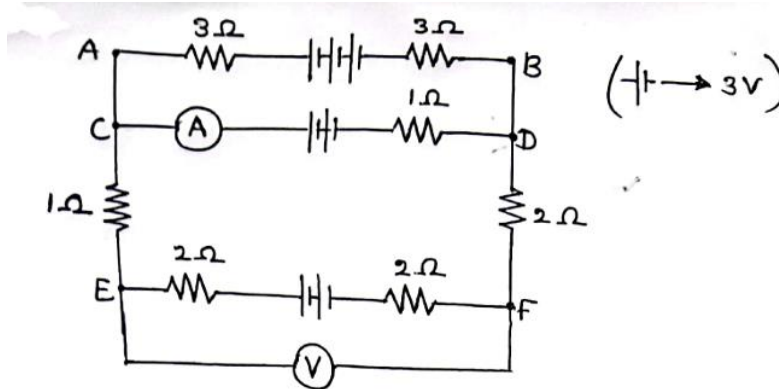


iii.

- There are seven 15 W electric bulbs, two 100 W electric fans a 1000 W electric iron, 2000 W refrigerator, 2500 W A/C machine in a luxury house. a person switches on each electric bulb and electric fans from 6:00 PM to 6:00 AM he uses the electric iron half an hour per day A/C machine six hours per day the refrigerator is switched on every time of every day. calculate the electric bill of the person for a month of 30 days. (1 kWh = Rs. 30: First 100 units, 1 kWh = Rs. 55: 100 - 300 units, 1 kWh = Rs. 95: 300 - 600 units, 1 kWh = Rs. 125: 600 - up units.)

- b. Due to high electric bill of the above-mentioned house, the owner hopes to fix solar cells. One solar panel generates one unit of electricity per day. National electricity board pays Rs. 43 for each extra generated electric unit. How many solar panels should be filled by the owner to obtain a profit equal to his previous electricity bill amount?

iv.



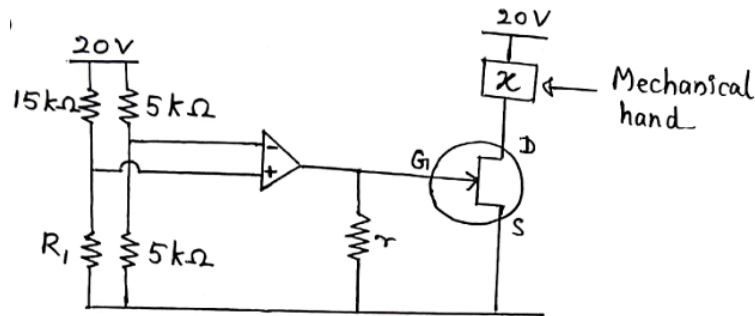
The initial resistance of each cell given is zero. Internal resistance of the ammeter is  $0.2\ \Omega$ . Voltmeter has a resistance of infinity. Electromotive force of each cell is  $3\text{ V}$ .

- Find the readings of the ammeter and the voltmeter.
- Find the supplied energy from the cells between A and B within 3 seconds.
- Find the amount of heat generated within the circuit during that time.
- If the answers are changed when considering between B and C state the reason for that.
- When the ammeter and the voltmeter is exchanged in the above circuit, give answers for above (a), (b) and (c).

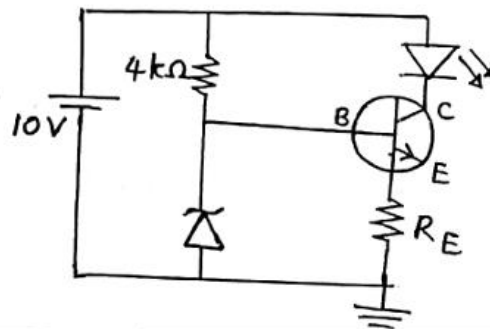
#### Part B

- Draw the n-channel FET transistor and p-channel FET transistor separately.
  - Write 2 similarities and 2 differences between n-channel FET transistor and npn transistor.
  - Draw the input and output characteristics of a n-channel FET transistor.
- In a milk powder production factory, the produced milk packets are sent to the packaging center through a conveyor belt. For the milk packet to be passed through the belt its mass should be  $600\text{ g}$ , when the mass of the milk packet is less than  $600\text{ g}$  it is removed from the strip by a mechanical hand. When the mass is greater than  $600\text{ g}$ , it damages the activity of the conveyor belt or its activity is stopped. The electric circuit used for this action is given below. the open loop gain of the op-amp is  $10^5$  and the saturated value of output is  $\pm 8\text{ V}$ , Here  $R'$  is a load sensitive resistor and the values of  $R'$  are  $15\ \Omega$ ,  $5\ \Omega$  and  $1\ \Omega$  at instances less than  $600\text{ g}$ ,  $600\text{ g}$  and greater than  $600\text{ g}$  respectively. here  $x$  represents the mechanical hand and for its operation there should be a  $2\text{ V}$  potential and  $5\text{ mA}$  current flowing through it. For  $I_D$  to become zero  $V_{GS}$  should be  $-8\text{ V}$ .

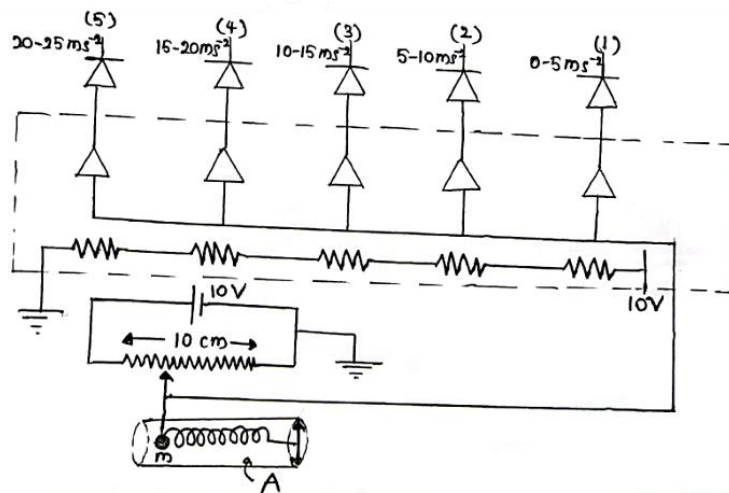
- Explain the reason for the decrease of the resistance in the load sensitive resistor when increasing the load.
- Calculate the output voltage of the operational amplifier when the load is 600 g, less than 600 g and greater than 600 g.
- Using the above answers, show that the mechanical hand activates only when the load is less than 600 g.
- If the mechanical hand works properly, calculate the resistance relevant for it.
- Calculate the value of  $V_{DS}$  when the mechanical hand is activated.



- c. A structure with a transistor used to supply a constant current at active state is given below. The EMF of this cell used here is 10 V and the breakdown voltage of Zener diode is 6 V. neglect the base current and calculate the followings,



- Find the current through the Zener diode.
  - Calculate the suitable resistance for  $R_E$  if 15 mA current passed through the LED at an instance ( $V_{BE} = 0.6 \text{ V}$ ).
  - If the voltage drop of the diode is 2.25 V at this instance, find the output voltage of the transistor.
- d. An accelerator produced by the students of the university of Moratuwa for the EXMO exhibition is given below. here the mechanical part is placed in the chamber A where m mass is fixed to the corner of a spring where the spring constant is K. there are five LED bulbs, and each LED represents different accelerations. Bulbs light up according to the range of accelerations. Ex: - when acceleration is  $14 \text{ ms}^{-2}$  the bulbs with (0-5), (5-10), (10-15) light up. In the op-amp used here, the supply voltage is  $\pm 10 \text{ V}$  and the saturated value of output is  $\pm 5 \text{ V}$ . The open loop gain is  $1 \times 10^5$ .



- i. Indicate how the operational amplifier should be connected by copying the parts with dotted lines and indicates the (+) and (-) poles.
- ii. What are the numbers of the bulb lighted up when the spring is,
  - (a) Compressed 2 cm
  - (b) Compressed 8 cm
- iii. If the vehicle exceeds  $20 \text{ ms}^{-2}$ , how does the acceleration recognize it.
- iv. At a instance, the vehicle 's acceleration was  $a$  and spring was compressed  $x$  distance. Write and expression for  $a$  using the terms  $m$ ,  $K$ , and  $x$ .
- v. If the acceleration of the vehicle is  $13.5 \text{ ms}^{-2}$  when the spring is compressed 5.4 cm, calculate the spring constant of the spring. ( $m = 0.5 \text{ kg}$ )

## 10. Part A

- i.
  - a. Define the zeroth law of thermodynamics.
  - b. Define Newton's Law of Cooling.
  - c. Define the specific latent heat of vaporization of water.
- ii.
  - a. The relative humidity of a room filled with air at a temperature of  $29^\circ\text{C}$  is 72% and its dew point is  $23^\circ\text{C}$ . If the room temperature is decreased by  $9^\circ\text{C}$ , by which fraction the water vapor inside the room will be condensed? The saturated vapor pressures of water at  $29^\circ\text{C}$  and  $20^\circ\text{C}$  are 35.7 mmHg and 20.17 mmHg respectively.
  - b. Plot a graph for the change of saturated and unsaturated vapor pressures with respect to time, for the above-mentioned process.
- iii. A man living in Antarctica accidentally forgets to close the cylindrical hot water flask. The thickness of the hot water flask is uniform throughout and the inner height of the flask without the lid is 15 cm while the external height is 18 cm. The heat conductivity of the material from which the bottle is made-up of is negligibly small. If the temperature is  $-13^\circ\text{C}$  and water temperature of the flask is at  $65^\circ\text{C}$  and if the water is filled up to the brim of the hot water flask,

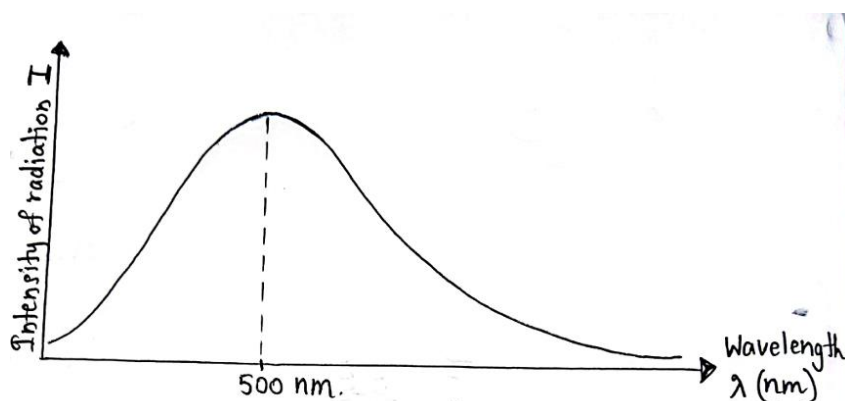
- a. Calculate how much of time is needed to completely freeze the water in the hot water flask (consider that all the water at  $0^{\circ}\text{C}$  converts into ice at  $0^{\circ}\text{C}$  completely before occurring any other temperature changes less than  $0^{\circ}\text{C}$  and water does not over falls during the whole process)
- b. If water overflows before it starts freezing calculate the volume of water over falls.

(density of ice =  $916.7 \times 10^{-3} \text{ gcm}^{-3}$ , density of water =  $1 \text{ gcm}^{-3}$ , heat conductivity of ice =  $2.22 \text{ Wm}^{-1}\text{K}^{-1}$ , surface emissivity of water = 0.98, specific heat capacity of water =  $4200 \text{ Jkg}^{-1}\text{K}^{-1}$ , surface emissivity of ice = 0.95, specific heat capacity of ice =  $2090 \text{ Jkg}^{-1}\text{K}^{-1}$ , internal diameter of the bottle = 7cm, specific latent heat of fusion of ice =  $3.34 \times 10^5 \text{ Jkg}^{-1}$ )

## Part B

Our sun is a star located about  $1.5 \times 10^8 \text{ km}$  away from earth, which can be also considered as a spherical black body which emits radiation primarily within three main wavelength regions. It emits radiations with wavelengths roughly between 400 and 700 nm in the visible light region, with shorter wavelengths in the ultraviolet region and with long wavelengths in the infrared region. Below is the graph of radiation intensity against the wavelength related to solar radiation.

(speed of light =  $3 \times 10^8 \text{ ms}^{-1}$ ,  $h = 6.6 \times 10^{-34} \text{ Js}$ ,  $\sigma = 5.7 \times 10^{-8} \text{ Wm}^{-2}\text{K}^{-4}$ , Wien's displacement constant (b) =  $2.9 \times 10^{-3} \text{ mK}$ )



- a.
  - i) What are the 3 main components of solar radiation?
  - ii) Define a blackbody.
  - iii) Plot two graphs of radiation intensity against wavelength on the same diagram for two identical black bodies at  $T_1$  and  $T_2$  temperatures, where  $T_1 > T_2$ . Mark the wavelengths with maximum intensity at temperatures  $T_1$  and  $T_2$  as  $\lambda_{m1}$  and  $\lambda_{m2}$  respectively.
  - iv) Write down the relationship between  $\lambda_{m1}$ ,  $\lambda_{m2}$ ,  $T_1$  and  $T_2$ .
  - v) What is the law used to derive the above relationship in (iv)?
- b.
  - i) Calculate the frequencies related to red and blue colors of visible light.
  - ii) Calculate the surface temperature of the sun.
  - iii) Assuming the sun to be a blackbody, calculate the radiation emitting intensity of the sun.

- iv) Then calculate the energy of solar radiation strikes a unit area near the Earth's atmosphere.
- v) If the mean wavelength of solar radiation is 550 nm, calculate the number of photons emitted per second by the sun.

c.

- i) The radiation energy calculated in the (iv) above, now strikes perpendicularly on to an object with an effective surface area of 5 m<sup>2</sup> on the earth. Considering the heat capacity of the object to be 400 JK<sup>-1</sup>, calculate its initial rate of increasing temperature.
- ii) Assuming the rate of increasing temperature calculated in (i) above is constant, find the temperature of the object after 50 s.
- iii) If the above object is exposed to solar radiation continuously for sometimes under practical conditions, plot the graph for the deflection of object's temperature against time. Give reasons for the specific shape of the graph obtained.



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